Solutions: Problem Set 1

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Problem 1.1 (Sampling distribution). Consider the population $U = \{1, 2, \ldots, 8\}$ of size $N = 8$. The values of the variable y are

```
y <- c(1, 2, 3, 3, 7, 8, 8, 8)
```

We are interested in the population total $t_y = \sum_{i \in U} y_i$. From a sample, estimate $\widehat{t}_y$ by $\widehat{t}_y = N\bar{y}$, where $\bar{y}$ is the sample mean.

a) Generate all simple random samples of size $n = 4$ from U without replacement. Denote the set of all such samples by $\mathscr{S}_4$. For each $s \in \mathscr{S}_4$, compute $\widehat{t}_y(s)$ and use the resulting values to obtain the sampling distribution of $\widehat{t}_y$.

Hint. Use the functions `combn()` and `choose()`.

Solution. Define the population and the sample size:

```
> N <- 8
> U <- 1:N
> y <- c(1, 2, 3, 3, 7, 8, 8, 8)
> n <- 4
```

Generate all samples:

```
> s4 <- combn(U, n)
> number_of_samples <- NCOL(s4)
> number_of_samples
```

```
[1] 70
```

Alternatively, the number of samples can be computed as follows:

```
> choose(N, n)
```

```
[1] 70
```

Compute the estimate for each sample:

```
> t_y <- apply(s4, 2, function(s) N * mean(y[s]))
```

Because all samples are equally likely, the sampling distribution is

```
> distr <- table(unname(t_y)) / number_of_samples
> distr
```

```
        18         26         28         30         32         36         38
0.01428571 0.02857143 0.10000000 0.05714286 0.04285714 0.04285714 0.12857143
        40         42         44         48         50         52         54
0.17142857 0.12857143 0.04285714 0.04285714 0.05714286 0.10000000 0.02857143
        62
0.01428571
```

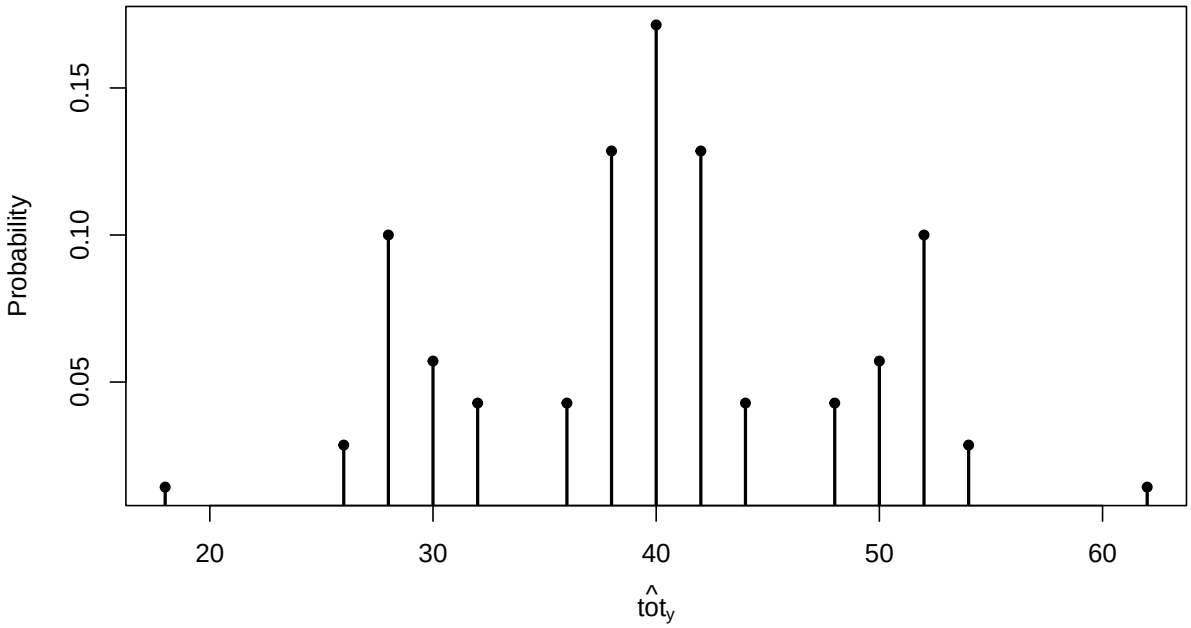


Figure 1: Sampling distribution of $\widehat{t}_y$

For example, $\mathbb{P}(\widehat{t}_y = 18) = 0.01428571$.

```
> tot <- as.numeric(names(distr))
> prob <- as.numeric(distr)
> plot(tot, prob, type = "h", lwd = 2, xlab = expression(hat(tot)[y]),
      ylab = "Probability")
> points(tot, prob, pch = 19, cex = 0.8)
```

b) Using the results from part a), compute numerically

$$\mathbb{E}(\widehat{t}_y) = \sum_{s \in \mathscr{S}_4} p(s)\, \widehat{t}_y(s)$$

and

$$\mathsf{var}(\widehat{t}_y) = \sum_{s \in \mathscr{S}_4} p(s) \left\{ \widehat{t}_y(s) - \mathbb{E}(\widehat{t}_y) \right\}^2,$$

where $p(s) = 1/\binom{N}{n}$.

Compare the numerical variance with the theoretical variance under simple random sampling without replacement:

$$\mathsf{var}(\widehat{t}_y) = N^2 \left(1 - \frac{n}{N}\right) \frac{\sigma_y^2}{n},$$

where

$$\sigma_y^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \bar{y}_U)^2.$$

Solution. Since all samples are equally likely,

```
> identical(mean(t_y), sum(y))
```

```
[1] TRUE
```

Hence $\mathbb{E}(\widehat{t}_y) = t_y$: the estimator is unbiased. The variance obtained from the sampling distribution is

```
> mean((t_y - mean(t_y))^2)
```

```
[1] 73.14286
```

The theoretical variance of $\widehat{t}_y$ is

```
> sigma2_y <- var(y)
> N^2 * (1 - n / N) * sigma2_y / n
```

```
[1] 73.14286
```

The two values are identical.

Problem 1.2 (Population Characteristics). We use the dataset `sepe.dat`, which contains data for a fictitious (sub-) population of firms:

```
> sepe <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/sepe.dat",
                     header = TRUE)
```

The dataset contains the number of full-time employees (`empl`), environmental protection expenditures (`expend`), and an indicator for industry class 31 (`code = 1`).

a) Compute the total environmental protection expenditures, the mean expenditures per firm, the expenditures per employee in the population, and the mean of the firm-level expenditures per employee.

Solution. We attach the data to the search space.

```
> attach(sepe)
```

The requested quantities are:

- Total expenditures:

  ```
  > sum(expend)
  ```

  ```
  [1] 68765270
  ```

- Mean expenditures per firm:

  ```
  > mean(expend)
  ```

  ```
  [1] 85956.59
  ```

- Expenditures per employee in the population:

  ```
  > sum(expend) / sum(empl)
  ```

```
[1] 446.0035
```

- Mean firm-level expenditures per employee:

```
> mean(expend[empl > 0] / empl[empl > 0])
[1] 1205.779
```

Remarks. Three of the 800 firms have no recorded full-time employees, so their firm-level expenditures per employee are undefined. If these zeros are treated as missing values, the last mean can be computed from firms with positive recorded employment. This implicitly assumes that the omitted firms would have the same average ratio as the observed firms.

b) Compute the population variance of the number of employees ${\sigma^2}_{\text{empl}}$, the population variance of expenditures ${\sigma^2}_{\text{expend}}$, and their correlation.

Solution. Since the dataset contains the full population, use the population variance with denominator N:

- Variance of the number of employees:

```
> mean((empl - mean(empl))^2)
[1] 36995.87
```

- Variance of expenditures:

```
> mean((expend - mean(expend))^2)
[1] 4632073291
```

- Correlation between the number of employees and expenditures:

```
> cor(empl, expend)
[1] 0.6597878
```

c) Compute the proportion P_{31} of firms in industry class 31.

Solution. Since `code` is a binary indicator, the proportion is its mean:

```
> mean(code)
[1] 0.1725
```

Problem 1.3 (Simulation). We continue to use the dataset `sepe.dat` (see Problem 1.2). We repeatedly draw simple random samples without replacement from the $N = 800$ firms.

a) For samples of size $n = 200$, use Monte Carlo simulation to estimate the variance of the estimator of mean environmental protection expenditures per firm (variable: `expend`). Draw 1,000 samples, compute the sample mean for each sample, and calculate the variance of the 1,000 estimates. Also create a histogram of the estimates.

Compare the Monte Carlo variance with the theoretical variance from Problem 1.2b) and report the coefficient of variation of the estimator.

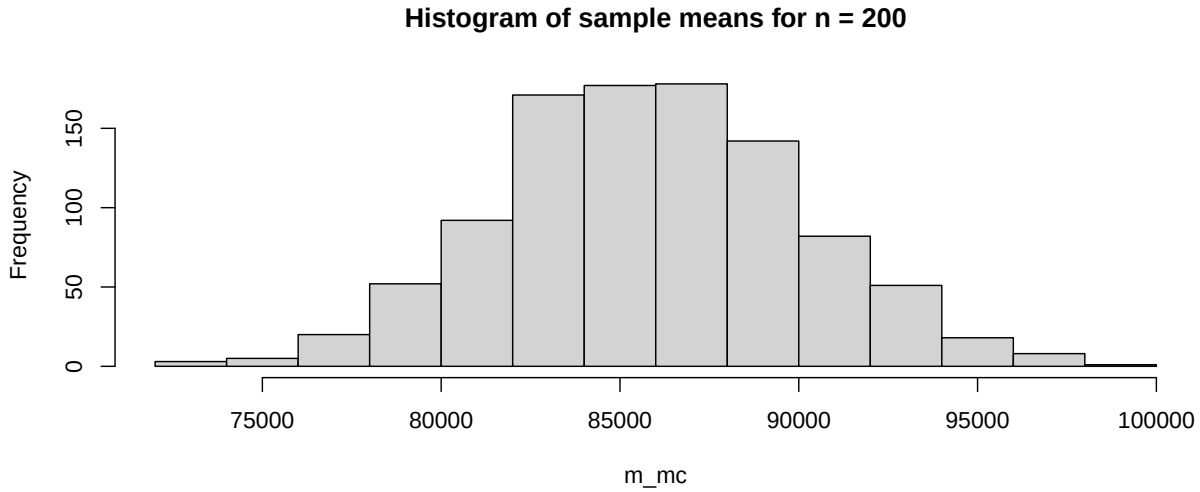


Figure 2: Simulated sampling distribution ($n = 200$)

Hint. Use `replicate(n_sim, expr)`, where `n_sim = 1000` and `expr` draws one sample and computes its sample mean. To reproduce the solution, use `set.seed(11)` before drawing the sample.

Solution. We use the dataset `sepe.dat`, which contains data for a fictitious population of firms:

```
> sepe <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/sepe.dat",
                     header = TRUE)
> attach(sepe)
```

Let `expend` be variable of interest (y-variable). Simulate 1,000 sample means:

```
> n_sim <- 1000
> n <- 200
> N <- 800
> set.seed(11)
> m_mc <- replicate(n_sim, mean(expend[sample(N, n)]))
```

The Monte Carlo variance is:

```
> VAR_m_mc <- var(m_mc)
> VAR_m_mc
```

```
[1] 17479711
```

Figure 2 shows the simulated sampling distribution.

```
> hist(m_mc, main = "Histogram of sample means for n = 200")
```

Under simple random sampling without replacement,

$$\mathsf{var}(\bar{y}) = \left(1 - \frac{n}{N}\right) \frac{\sigma_y^2}{n},$$

where

$$\sigma_y^2 = \frac{1}{N-1}\sum_{i \in U}(y_i - \bar{y}_U)^2.$$

In R, `var(expend)` computes σ_y^2. The theoretical variance is:

```
> VAR_m_y <- (1 - n / N) * var(expend) / n
> VAR_m_y
```

```
[1] 17392015
```

The theoretical and Monte Carlo coefficients of variation are:

```
> sqrt(VAR_m_y) / mean(expend)
```

```
[1] 0.04851721
```

```
> sd(m_mc) / mean(m_mc)
```

```
[1] 0.04870305
```

The Monte Carlo results are close to the theoretical values.

b) Repeat the simulation for samples of size $n = 50$. Compare the coefficient of variation with the result from part a). Also create a histogram of the estimates.

Solution. Repeat the simulation with $n = 50$:

```
> n <- 50
> set.seed(11)
> m_mc50 <- replicate(n_sim, mean(expend[sample(N, n)]))
```

The Monte Carlo variance is:

```
> VAR_m_mc <- var(m_mc50)
> VAR_m_mc
```

```
[1] 79657236
```

Figure 3 shows the simulated sampling distribution.

```
> layout(matrix(c(1, 2), ncol = 1))
> hist(m_mc, xlim = c(6, 13) * 1E4,
       main = "Histogram of sample means for n = 200")
> hist(m_mc50, xlim = c(6, 13) * 1E4,
       main = "Histogram of sample means for n = 50")
```

The theoretical variance is:

```
> VAR_m_y <- (1 - n / N) * var(expend) / n
> VAR_m_y
```

```
[1] 86960074
```

The theoretical and Monte Carlo coefficients of variation are:

```
> sqrt(VAR_m_y) / mean(expend)
```

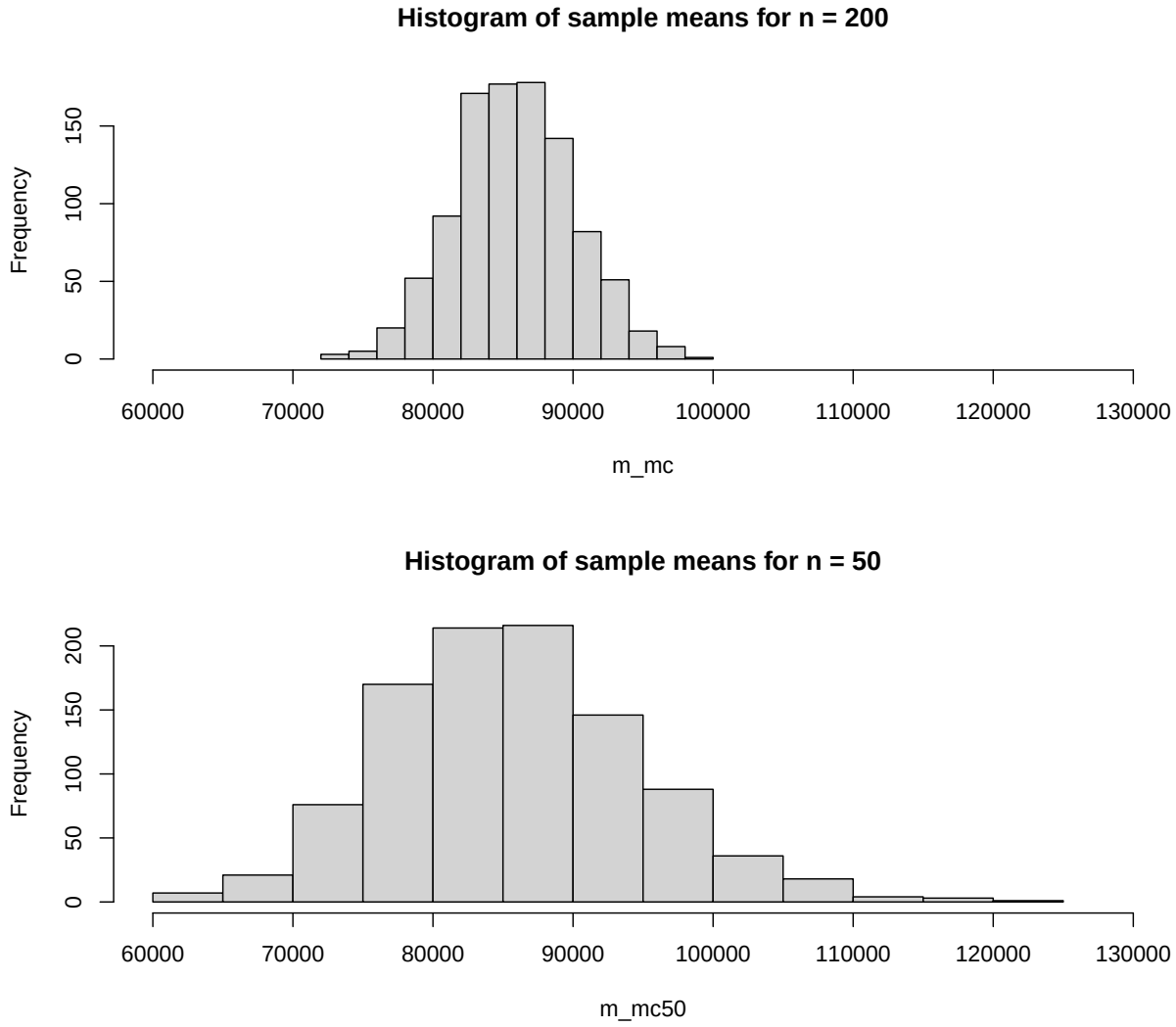


Figure 3: Simulated sampling distribution ($n = 50$ and $n = 200$; x-axis uses the same scale)

```
[1] 0.1084878
```

```
> sd(m_mc50) / mean(m_mc50)
```

```
[1] 0.1040796
```

With the smaller sample, both the variance and the coefficient of variation are larger, as expected.